

PERFORMANCE TESTING

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
Team Members	G Uday Kumar, Kavali Anu, Pavan Kumar Sowdepalli, Dasarayyagari Sravan Kumar
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance testing evaluates how your system behaves under expected and peak load conditions[cite: 22]. Complete the sections below to document your testing approach, tools used, and results obtained[cite: 22].

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter[cite: 22]
Type of Testing	Load Testing, Stress Testing, Spike Testing[cite: 22]
Target Module / API	Crop Recommendation and Optimization API (/predict)[cite: 22]
Test Environment	Localhost (Python Flask Server on Local Machine)[cite: 22]
Test Date	03 July 2026[cite: 22]

Step 2: Test Scenarios

S.No	Test Scenario	Description	No. of Virtual Users	Duration (seconds)
1	Normal Load Test	Multiple farmers and operators accessing prediction API concurrently[cite: 22].	10[cite: 22]	60[cite: 22]
2	High Load Test	Simulate heavy concurrent regional analytics requests[cite: 22].	50[cite: 22]	120[cite: 22]
3	Stress Test	Push the agricultural model recommendation capacity beyond normal operational capacity[cite: 22].	100[cite: 22]	180[cite: 22]
4	Spike Test	Sudden intense increase in user load to mimic harvest season usage spikes[cite: 22].	200[cite: 22]	60[cite: 22]